LAB 4

UML Activity & Sequence Diagram Modeling

Digital Health Record Management System (DHRMS)

Complete Lab Report • Requirements • Use Cases • Rendered UML Diagrams

Based on: DHRMS_Complete_SRS_v2.0.pdf

Name: _____

Roll Number: _____

Branch / Section: _____

Submitted to: _____

Date: _____

1. Aim

To model the major workflows of the Digital Health Record Management System (DHRMS) using UML Activity Diagrams and UML Sequence Diagrams, while documenting the actors, functional requirements, security constraints, and use-case preconditions and postconditions.

2. Objectives

• Represent major system workflows, decisions, and alternate/error paths.

• Show interactions and message order between actors and system components.

• Document actor responsibilities and use-case functions.

• Reflect authentication, role-based authorization, consent checks, secure document handling, interoperability, and audit logging described in the SRS.

• Keep access to confidential medical information restricted, traceable, and protected.

3. Project Overview

DHRMS is a centralized, secure web- and service-based platform that manages patient electronic health records, diagnostic reports, treatment histories, prescriptions, clinical encounters, consent records, and audit information. It supports patients, healthcare providers, system administrators, compliance/auditors, and authorized external healthcare systems. The system separates identity management, authorization, clinical data, consent, document management, interoperability, and audit logging into logical subsystems.

4. Actors and Responsibilities

Actor	Responsibilities
Patient	View personal health records, diagnostic reports, prescriptions, treatment history and encounters; manage profile; grant, rev
Healthcare Provider	Search for patients; access only authorized records; create clinical encounters; enter vitals and diagnoses; manage diagnos
System Administrator	Manage accounts, roles, permissions and system configuration; monitor operational and security events.
Compliance / Auditor	Review immutable audit records; investigate security events; produce compliance reports when authorized.
External Lab / Hospital	Exchange authorized structured healthcare data with DHRMS through secure, authenticated interoperability interfaces.

5. Functional Requirements Summary

ID / Area	Requirement summary
Authentication	Registered users can authenticate and log out; credentials must be validated; every session maps to a unique identity; passwords are not
Patient health records	Maintain patient profiles and longitudinal records; retrieve and modify records only through authorized workflows; link records to the correc
Role-based access control	Enforce roles and granular permissions; deny unauthenticated or unauthorized requests; isolate administrative privileges.
Consent management	Allow patients to grant, modify, review and revoke provider access; enforce scope restrictions and validity/expiry; revoked consent must bl
Clinical encounters	Document provider-patient encounters, vitals, clinical notes, diagnoses and assessments.
Diagnostic reports	Create and manage diagnostic reports, metadata, coding such as LOINC where applicable, and related clinical attachments.
Treatment and prescriptions	Document treatment plans, author digital prescriptions, and track medication details.
Medical documents	Validate file MIME/type and metadata, calculate SHA-256 integrity hash, and store documents securely in encrypted object storage.
Audit and compliance	Record access to sensitive records and authorization failures in a chronological, immutable/append-only audit trail.
Interoperability	Support structured healthcare exchange using HL7 FHIR Release 5 (R5) resources, through secure authenticated external connections.
Administration / monitoring	Provide role-specific dashboards, user/permission management, system configuration, and security-event monitoring.
Communications	Use HTTPS/TLS for protected communications; use JSON for typical REST requests/responses; return controlled errors without leaking in

6. Non-Functional and Security Requirements

Category	Requirement / design constraint
Confidentiality	Treat healthcare information as confidential; expose only data authorized for the current user and purpose.
Authentication & authorization	Protected resources require authentication and permission checks; MFA is supported; sensitive actions require confirmation where appl
Transport security	Protected browser-server communication uses HTTPS/TLS; medical information must not travel over unencrypted channels.
Password safety	Never store passwords in plaintext; use secure password verification and generic invalid-credential errors.
Integrity	Use SHA-256 to identify document integrity; validate/sanitize external healthcare data before processing.
Auditability	Log sensitive record access, consent changes, security events and authorization failures; audit trail is immutable/append-only.
Interoperability	Use HL7 FHIR R5 representations where required; authenticate external connections using secure mechanisms such as mTLS/OAuth2
Usability	Provide role-specific navigation, clear validation errors, and controlled authorization failures.
Operating environment	Modern desktop/mobile browsers; containerized Linux server environment; PostgreSQL 16, Redis 7 and S3-compatible encrypted objec

7. UML Use Case Diagram

The overview below connects the five actor groups to the major DHRMS use cases. It is a high-level use-case view; detailed use-case specifications follow.

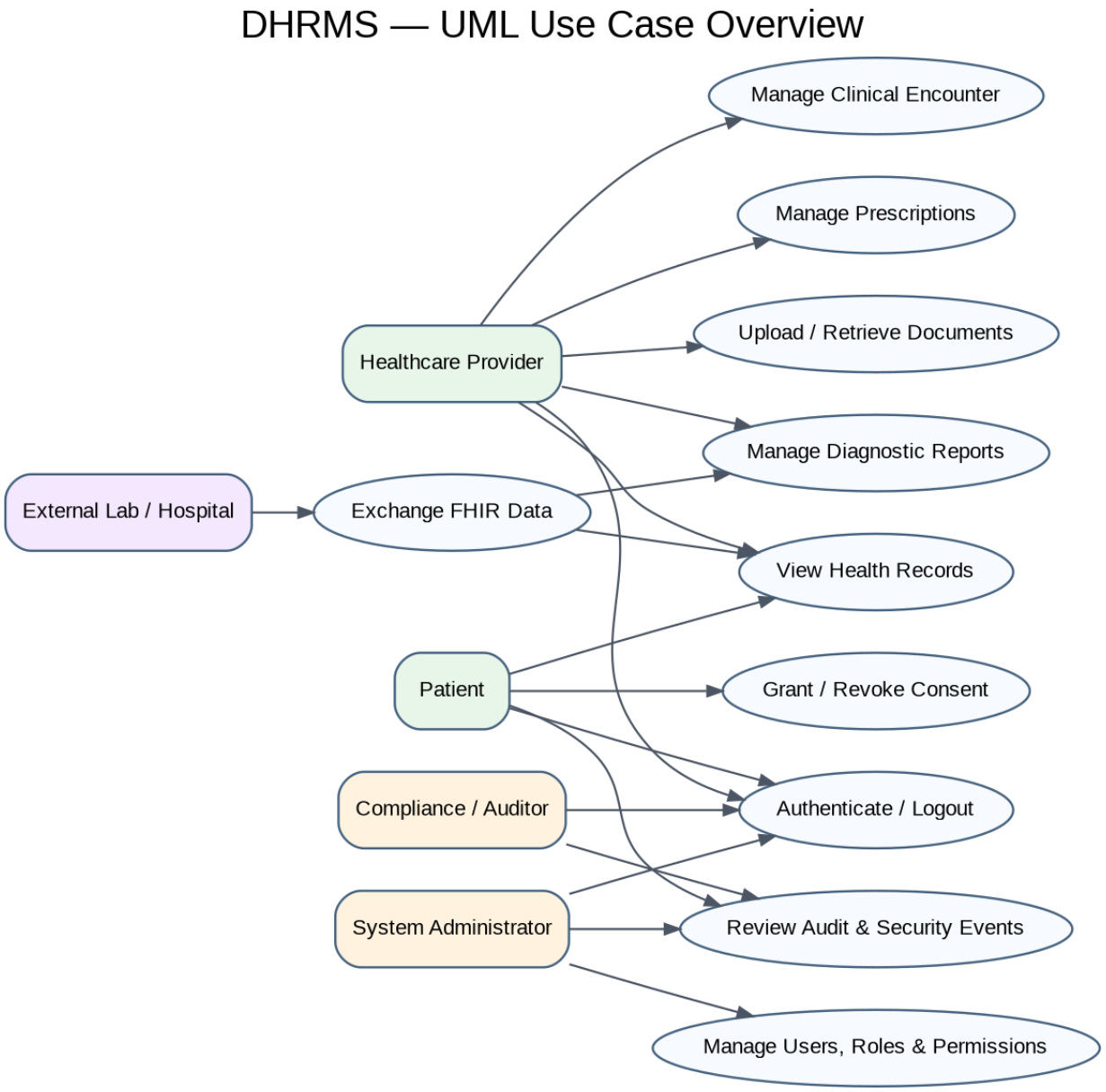


Figure 1. DHRMS UML use case overview.

8. Major Use Cases: Function, Preconditions and Postconditions

Use case	Primary actor(s)	Preconditions	Postconditions / result
Authenticate / Logout	All interactive users	User has an account; login service is available.	Valid user receives authenticated session and role permissions; invalid a
View patient records	Patient; Healthcare Provider	Requester is authenticated; provider access also requires role permission and active consent; patient is selected.	Only permitted records are returned; access event is audited; denied atte
Manage consent	Patient	Patient is authenticated and authorized to manage their consent records.	Consent is created/updated/revoked with scope and validity; change is a
Manage clinical encounter	Healthcare Provider	Provider authenticated; patient selected; role and applicable consent checks pass.	Validated encounter data is stored and linked to patient/provider; audit e
Manage prescription	Healthcare Provider	Provider authorized for the patient and encounter; medication details are available.	Validated prescription is stored and made available through permitted pa
Manage diagnostic report	Healthcare Provider; authorized external system	Authenticated/authorized user or verified external connection and report data are available.	Report and metadata are stored/linked; interoperability or report action is
Upload medical document	Healthcare Provider / authorized user	Authenticated user has permission to upload for the target patient.	File passes type/validation checks; hash and metadata are stored; file is
Manage users and roles	System Administrator	Administrator is authenticated and has administrative privilege.	Approved role/permission changes are applied and auditable.
Review audit trail	Compliance / Auditor; authorized administrator	User is authenticated and has audit-review permission.	Authorized audit entries/security events are reviewed; audit records rem
Exchange FHIR data	External Lab / Hospital	External system uses an authenticated secure connection and FHIR resource is accepted/processed.	Valid FHIR exchange is completed; invalid data is rejected; exc

9. Activity Diagram 1 — Authentication and MFA

Workflow: the user submits credentials; the system verifies the stored password hash and, if configured, verifies a one-time MFA code. A valid user receives a session and signed token and sees a role-specific dashboard. Invalid authentication is rejected, logged, and may result in account lockout after five consecutive failures where that policy is configured.

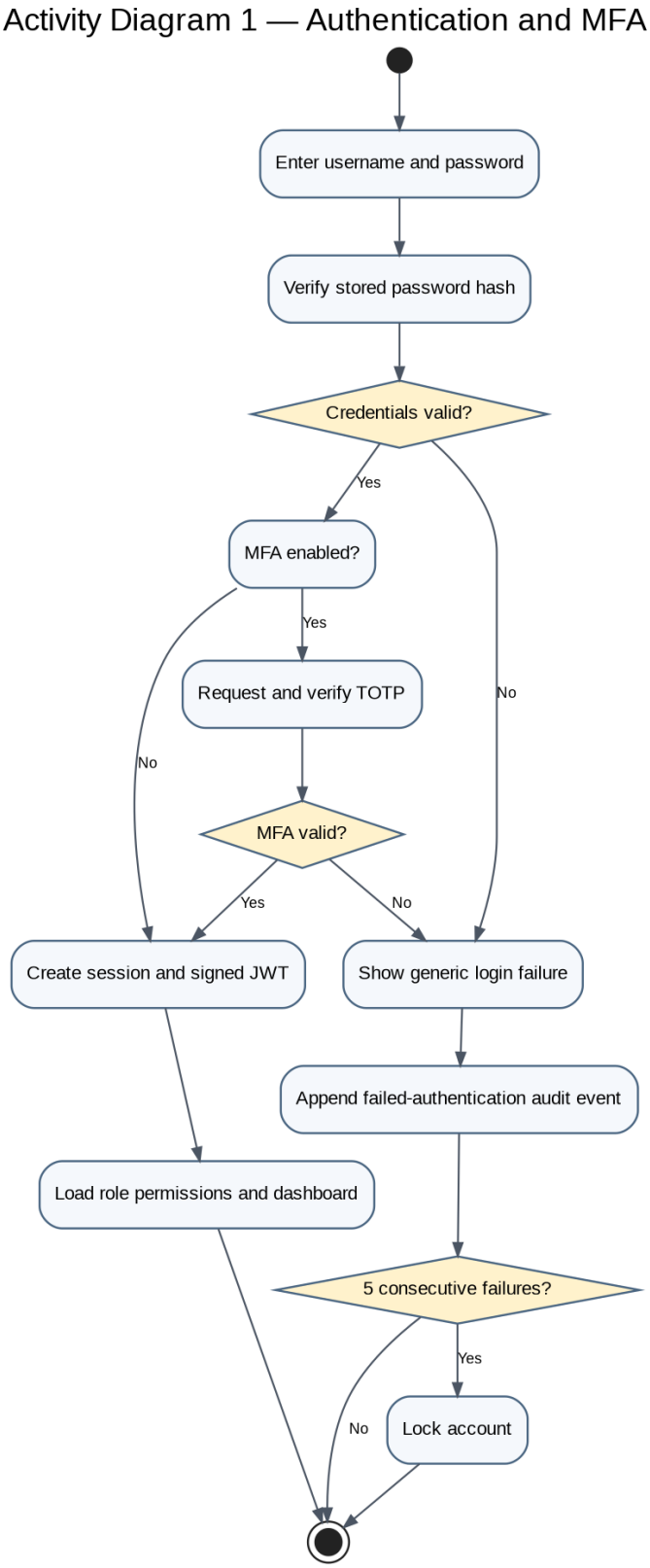


Figure 2. Authentication activity workflow.

10. Activity Diagram 2 — Patient Health Record Retrieval

Workflow: DHRMS verifies identity, role permissions, active consent, and requested data scope before retrieving records. Any failed check denies the request. Successful access and authorization failures are written to the audit trail.

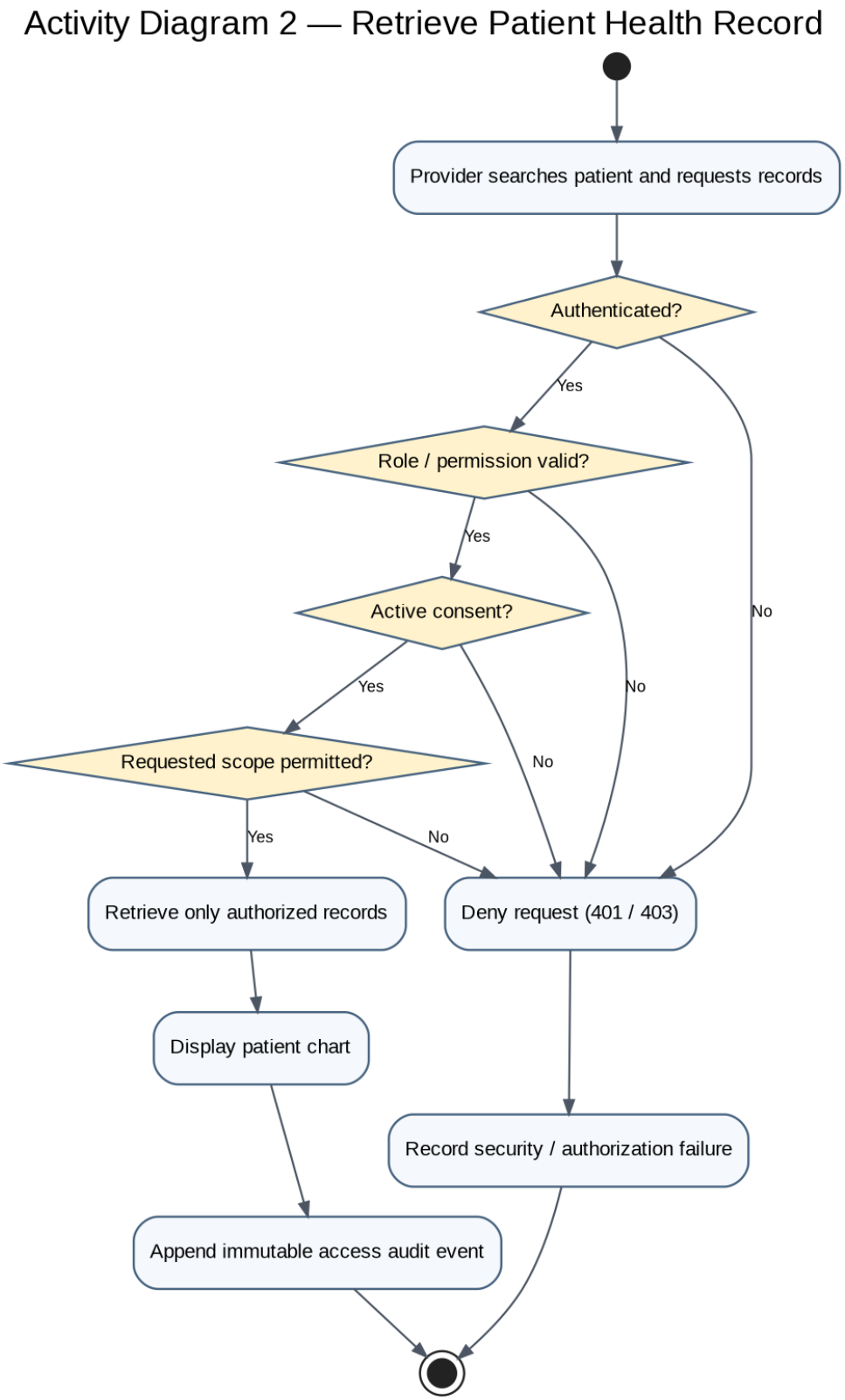


Figure 3. Patient record retrieval activity workflow.

11. Activity Diagram 3 — Consent Lifecycle

Workflow: a patient may grant consent by selecting a provider, access scope and validity; modify an existing directive; or revoke it. Consent state changes are audited. A revoked directive must prevent future access covered by that consent.

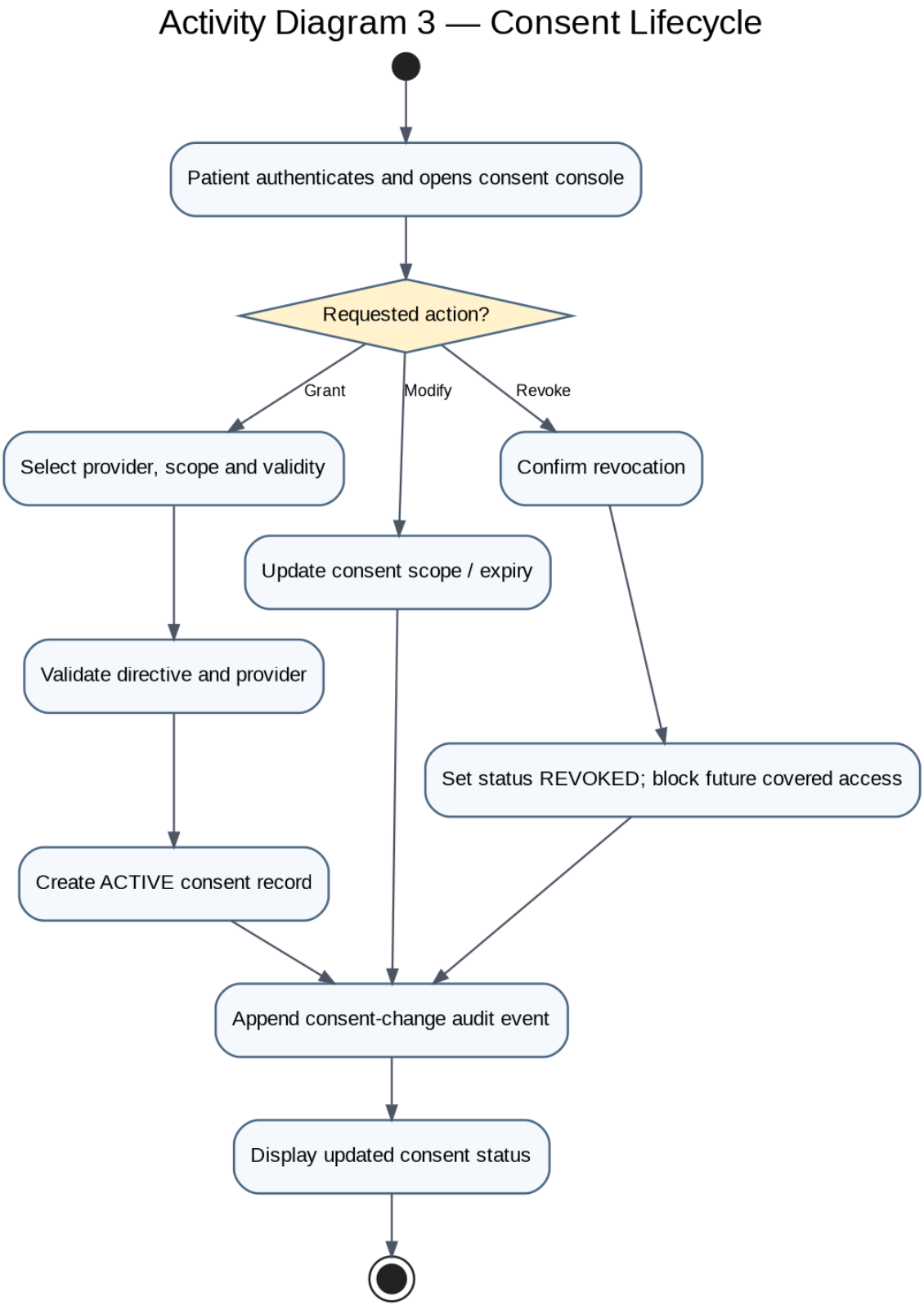


Figure 4. Consent lifecycle activity workflow.

12. Activity Diagram 4 — Clinical Encounter and Prescription

Workflow: an authorized provider selects a patient and passes role/consent checks, enters encounter information, vitals, notes and diagnosis, and validates the data. The encounter is stored and audited. If a prescription is required, medication, dosage, frequency and duration are validated before saving.

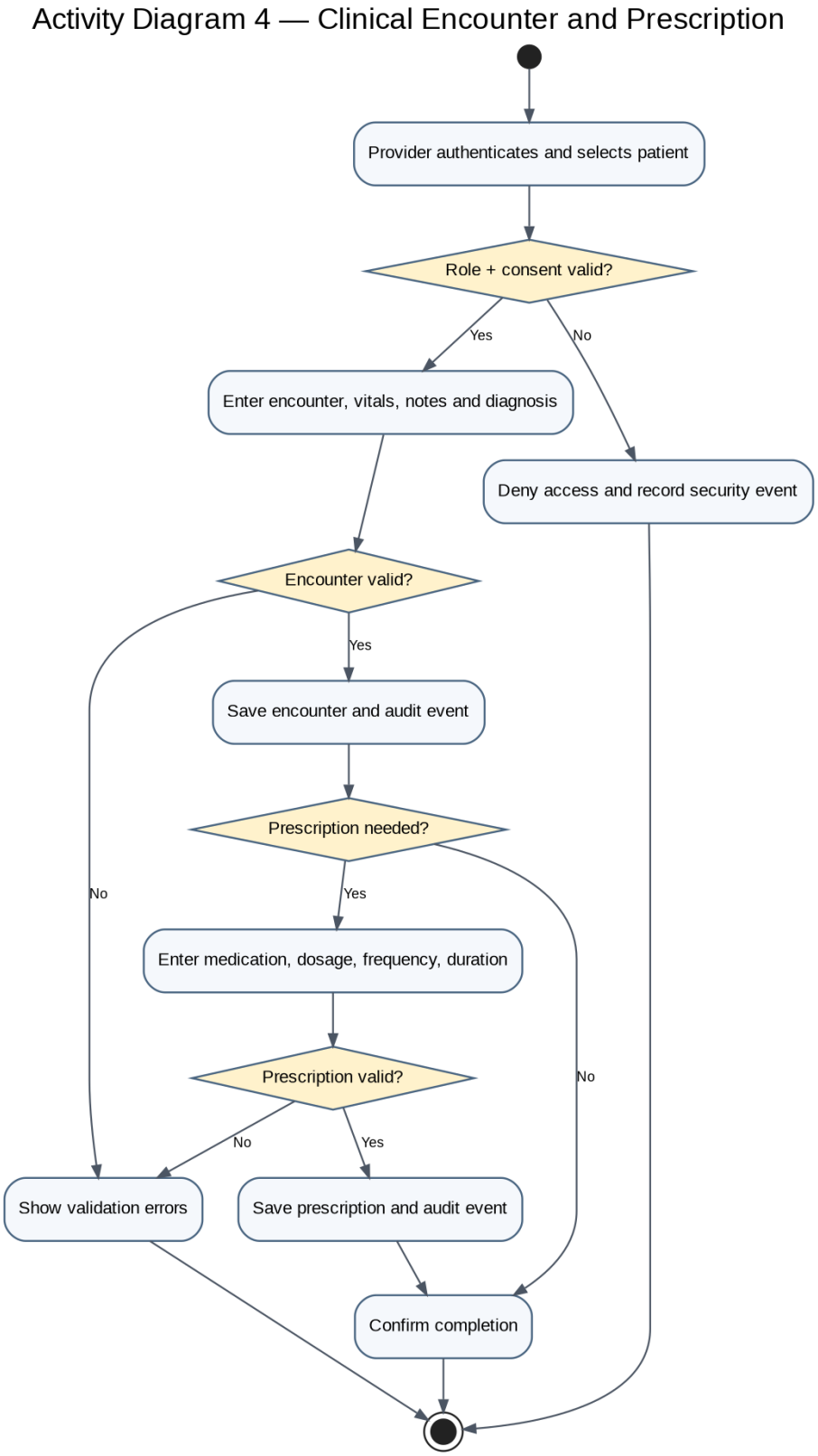


Figure 5. Clinical encounter and prescription workflow.

13. Activity Diagram 5 — Secure Medical Document Upload

Workflow: the system verifies user permissions, validates the file type and metadata, calculates a SHA-256 integrity hash, stores the file in encrypted object storage, persists metadata, and appends an audit event. Invalid or unauthorized uploads are rejected.

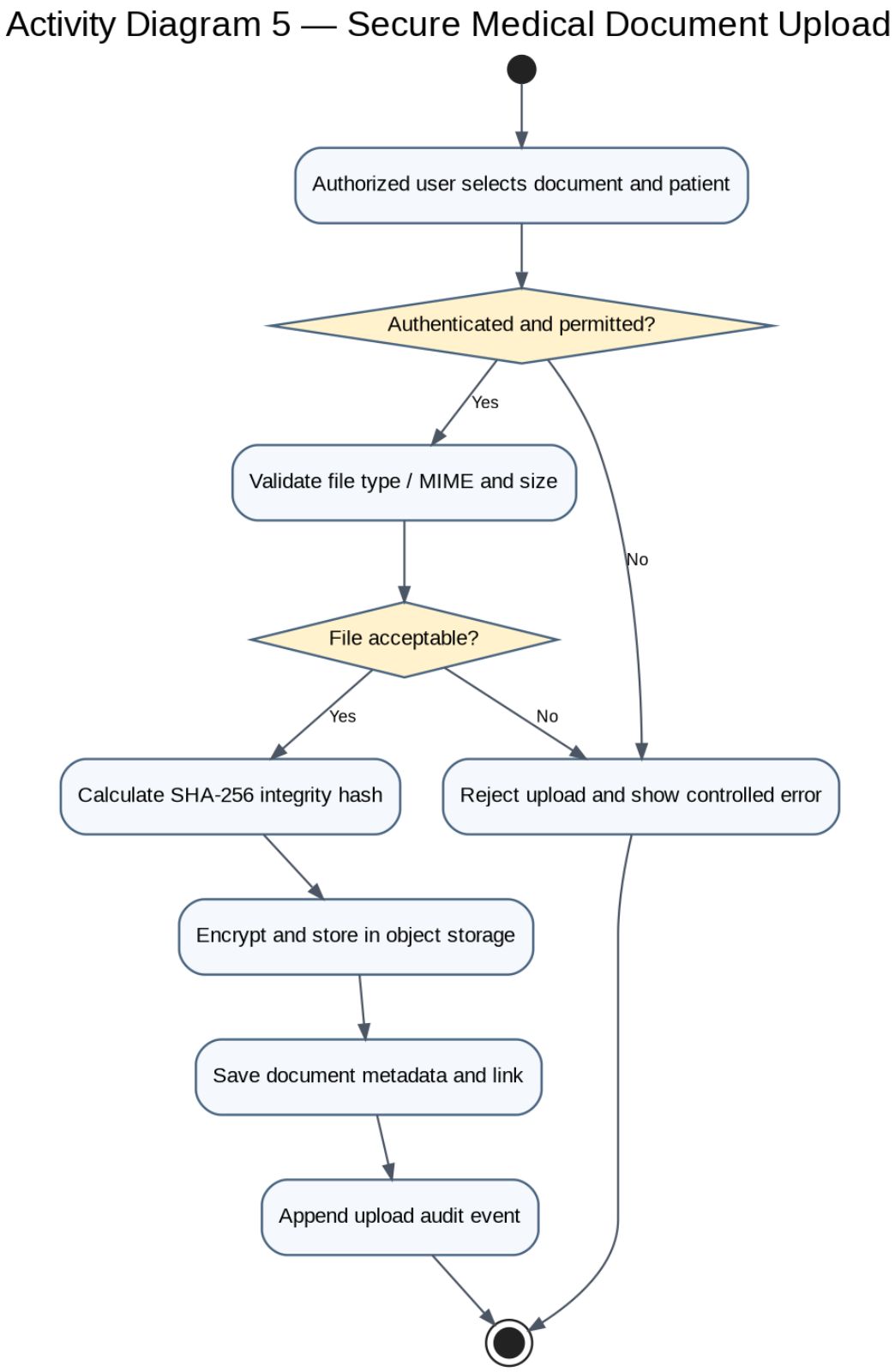


Figure 6. Secure medical-document upload workflow.

14. Sequence Diagram 1 — Authentication

The sequence illustrates the order of messages between the user, portal, authentication service, user database and audit service. The system records the authentication outcome and returns either an authenticated session or a generic failure response.

Sequence Diagram 1 — Authentication



Figure 7. Authentication message sequence.

15. Sequence Diagram 2 — Patient Record Retrieval

The API coordinates token and role validation, consent/scope verification, clinical data retrieval, and immutable audit logging. The system returns only the records that the requester is allowed to access.

Sequence Diagram 2 — Authorized Record Retrieval

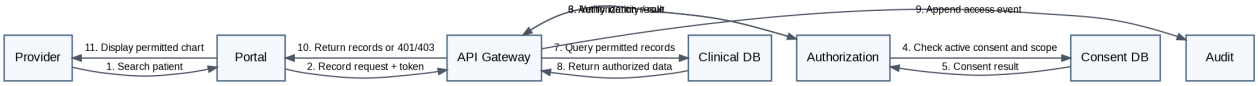


Figure 8. Authorized patient-record retrieval sequence.

16. Sequence Diagram 3 — External FHIR Data Exchange

An external laboratory or hospital sends a FHIR R5 resource. DHRMS authenticates the connection, validates and sanitizes the resource, processes valid data, logs the exchange, and returns a controlled response.

Sequence Diagram 3 — External FHIR Data Exchange

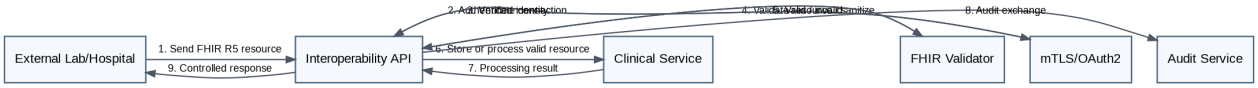


Figure 9. External FHIR exchange sequence.

17. Design Assumptions and Clarifications

• These diagrams are a workflow model derived from the DHRMS SRS, not a claim that a deployed system already exists.

• HTTP 401 is shown for unauthenticated requests and HTTP 403 for authenticated requests that lack permission; exact API status mapping may depend on implementation.

• Five-failure account lockout is shown as a design workflow where configured; retain the exact threshold/policy specified by the version of the SRS or implementation being assessed.

• mTLS/OAuth2 are shown as secure external-connection mechanisms for interoperability; exact protocol configuration depends on the integration agreement.

• All protected requests and sensitive medical data must use secure transport, and audit records must not be silently altered.

18. Result

The DHRMS project was modeled using one UML Use Case Diagram, five UML Activity Diagrams, and three UML Sequence Diagrams. The report also documents actor responsibilities, functional and non-functional requirements, and preconditions/postconditions for major use cases. The diagrams cover authentication, patient-record access, consent lifecycle, clinical encounters and prescriptions, secure medical-document upload, and external FHIR data exchange.

19. Tool and Submission Notes

• Diagrams are rendered directly in this PDF and are not just PlantUML source-code listings.

• To edit the diagrams, recreate them in diagrams.net (Draw.io) using UML actor, use-case, activity, decision, lifeline and message shapes.

- Suggested tool: https://app.diagrams.net/
- Before submission, fill in your name, roll number, section, instructor and date.